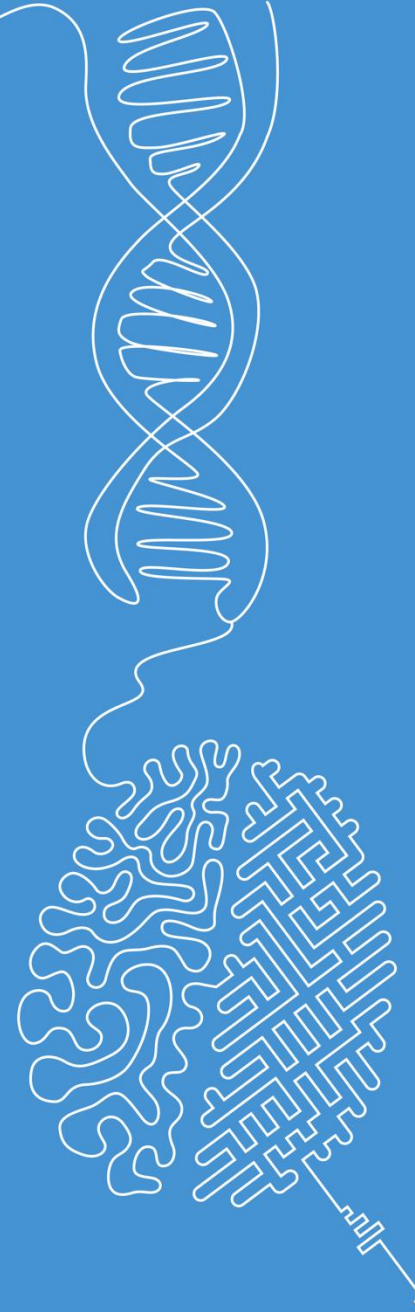


Sampling

Image and Signal Processing

Norman Juchler

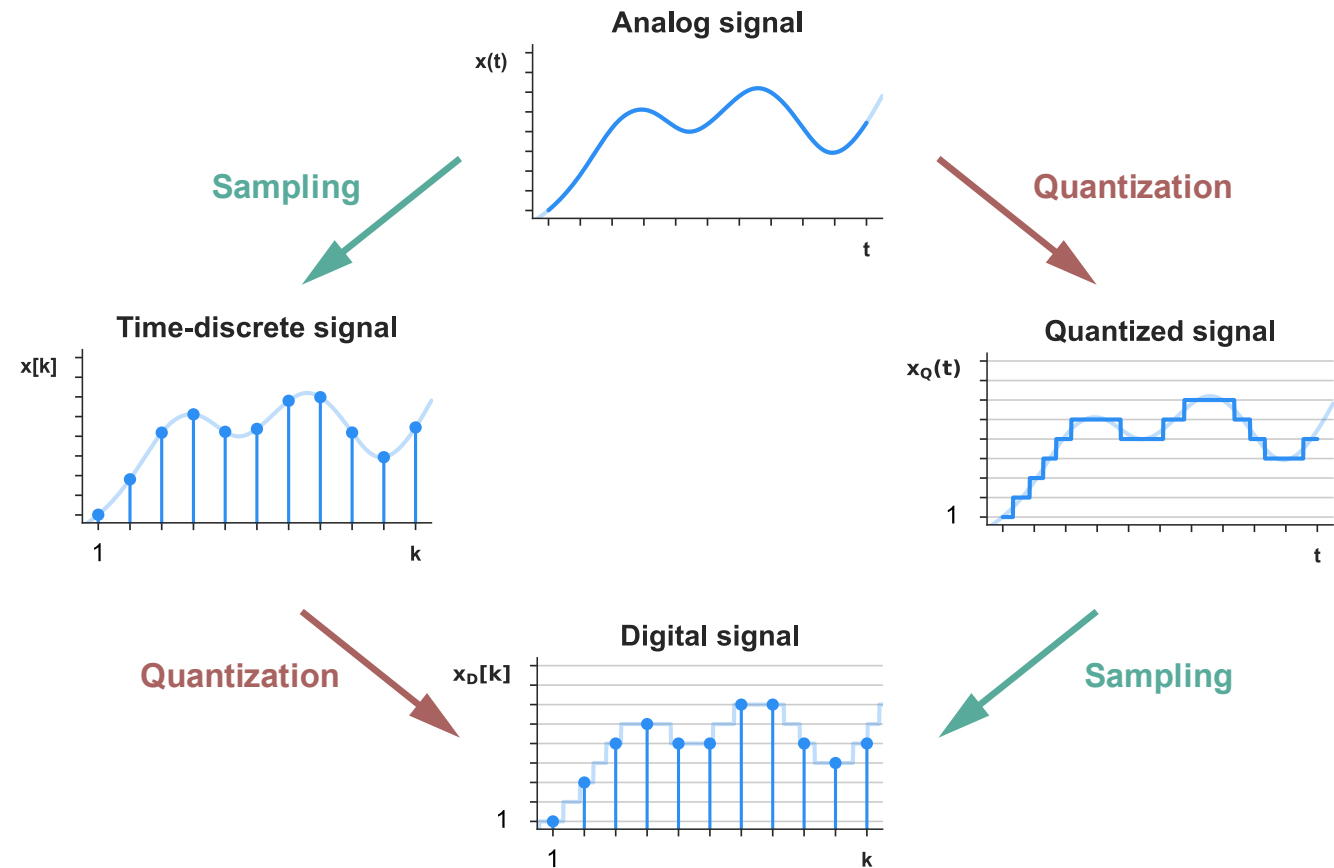


Analog to digital conversion

From a continuous to a digital signal

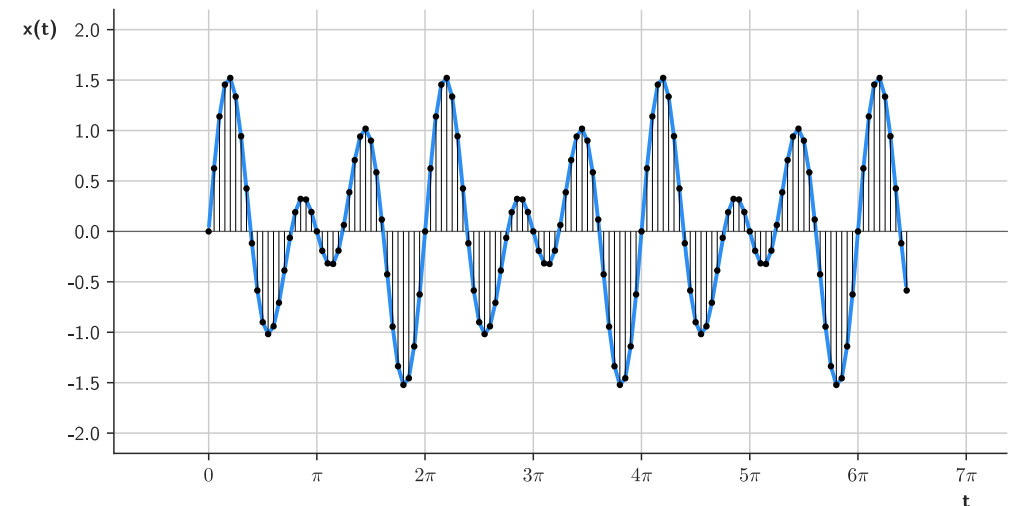
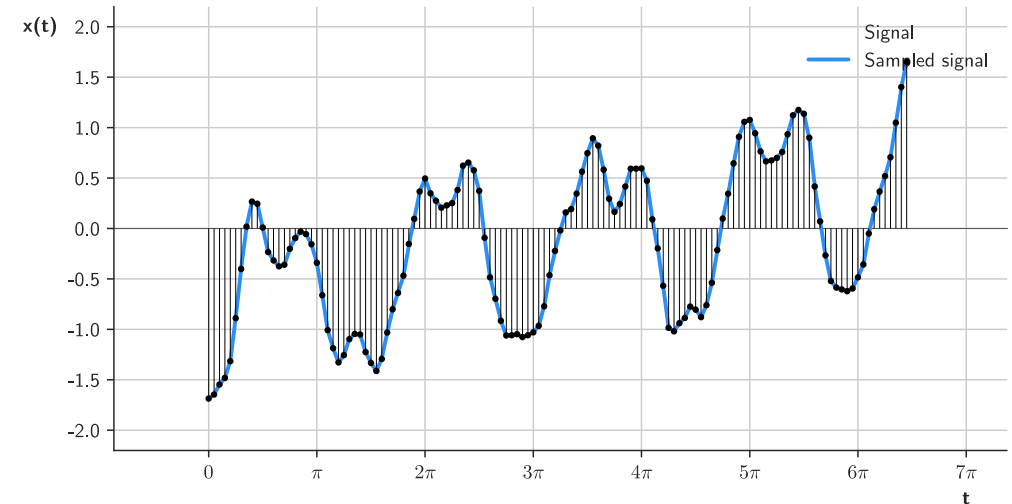
Sampling vs. quantization

- **Sampling** converts a continuous-time signal into a discrete-time signal by measuring it at regular time intervals.
- **Quantization** approximates continuous amplitudes using a finite set of values (e.g. integers).
- These two steps form the basis for digitizing analog signals.
- Quantization error: The difference between the original signal and the quantized signal (rounding error)



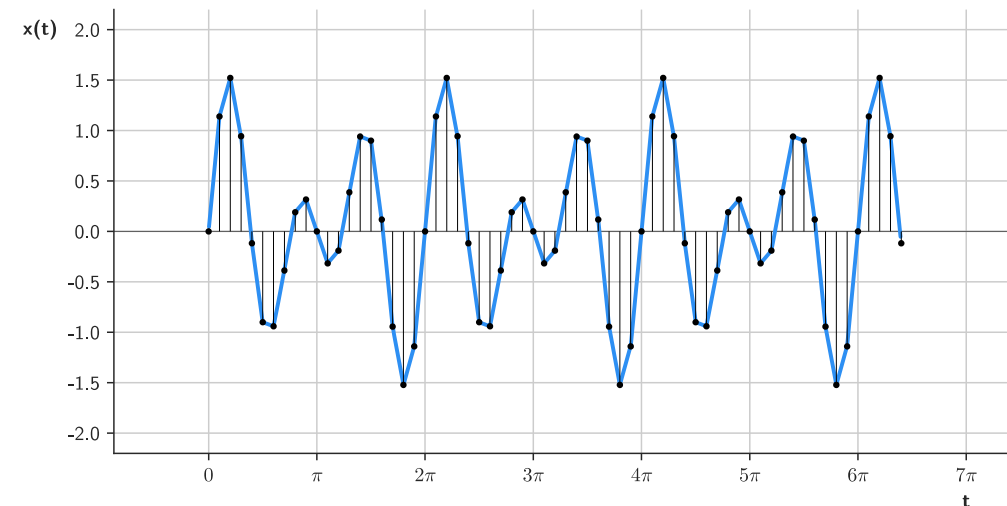
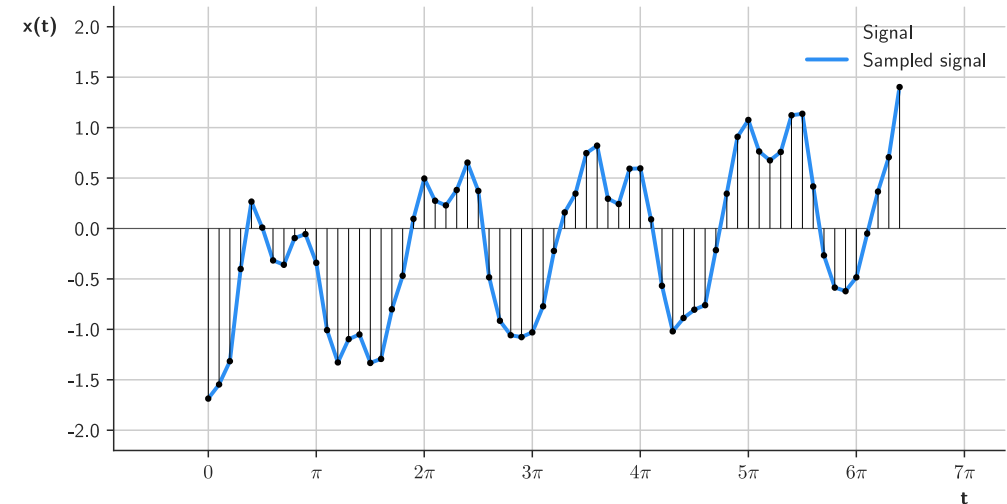
Over- and undersampling

- **Sampling objective:** Capture the fundamental features of a signal without significantly compromising its characteristics due to the sampling process.
- The adjustable parameter is the **sampling rate** $f_s = 1/T_s$. We must strike a balance between capturing sufficient signal information and minimizing data storage and processing requirements.
- Situation on the right: **oversampling**. Everything is great.



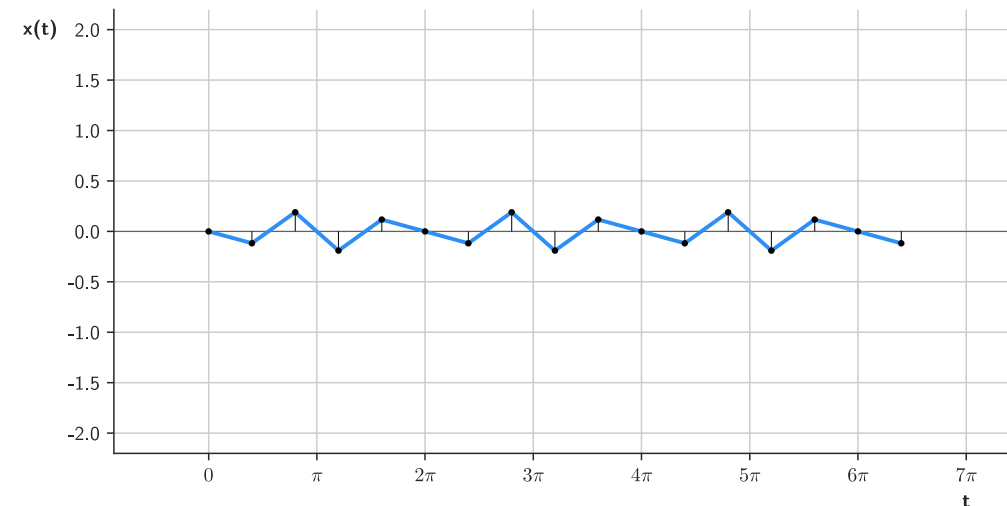
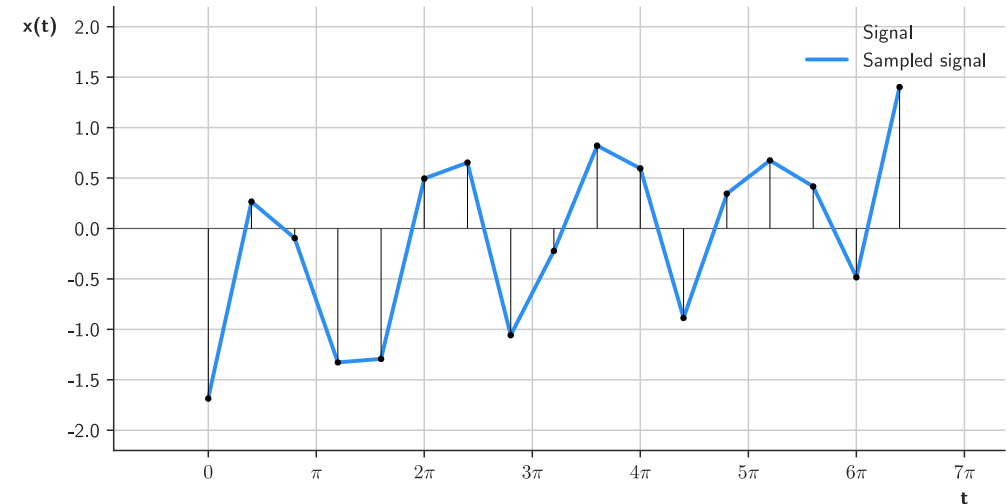
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- The adjustable parameter is the **sampling rate** $f_s = 1/T_s$. We must strike a balance between capturing sufficient signal information and minimizing data storage and processing requirements.
- Situation on the right: **oversampling**. Everything looks okay.



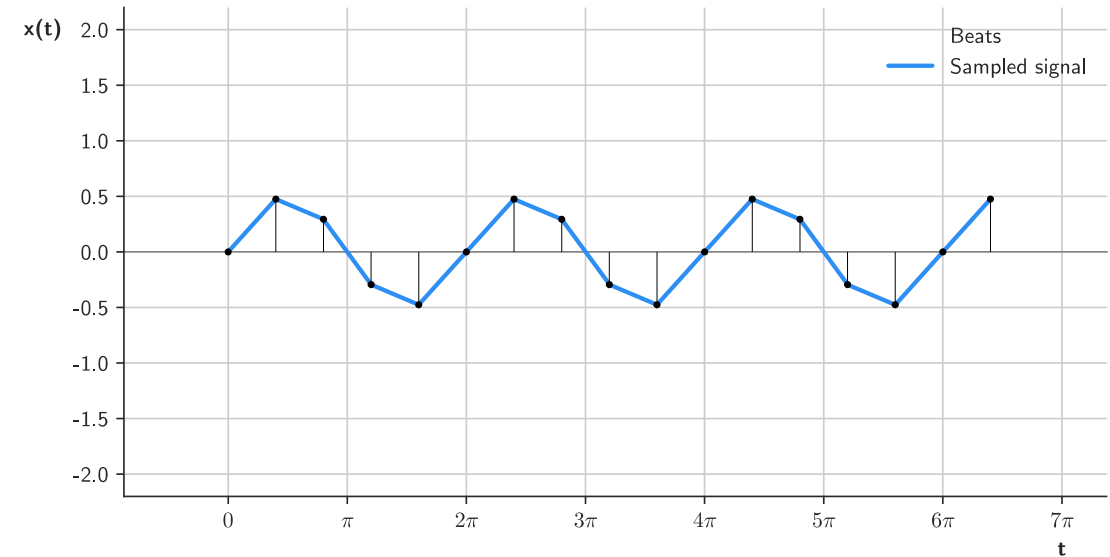
Over- and undersampling

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- The adjustable parameter is the **sampling rate** $f_s = 1/T_s$. We must strike a balance between capturing sufficient signal information and minimizing data storage and processing requirements.
- Situation on the right: **undersampling**. We have lost substantial information because the sampling rate is too small for our data.



Effects of undersampling: Loss of information and aliasing

- Undersampling occurs when the sampling rate is too low for a given signal or when the signal contains frequencies too high for the chosen sampling rate.
- This not only leads to **information loss** but can also introduce artificial frequencies (**aliasing**) that were not present in the original signal.
- Avoidance strategy: Low-pass filtering
- Videos illustrating aliasing:
 - Illustration of aliasing: [YouTube](#) (by Gergely)
 - A bit more technical: [YouTube](#) (by Zach Star)
(No worries if you don't understand everything)



Nyquist–Shannon sampling theorem

- What is the natural boundary between over- and undersampling?
- The **Nyquist-Shannon sampling theorem** provides the answer:

The sample rate must be at least twice the maximal frequency of the signal to avoid the effects of undersampling: $f_s > 2f_{\max}$

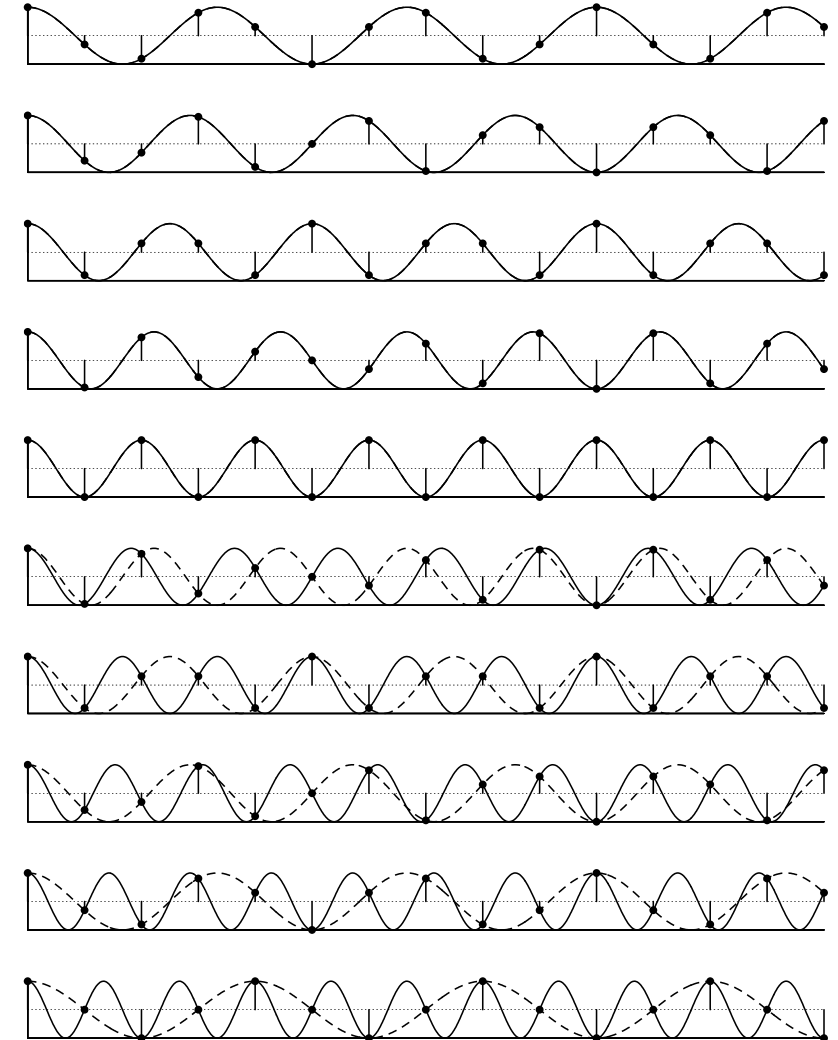
- The theorem is a fundamental principle in digital signal processing!
- Alternative formulation: If a function $x(t)$ contains no frequencies higher than $f_{\max}$ Hertz, then it can be completely determined from sampling points spaced (strictly) less than $T_s < \frac{1}{2f_{\max}}$.

Sampling across the critical frequency

- The frequency of a sampled signal is increased from top to bottom, while the sampling frequency is constant.
- The dashed lines are possible aliases that would have the same samples.

Tasks:

- Describe in your own words what happens.
- Which illustration shows the situation at the critical frequency?



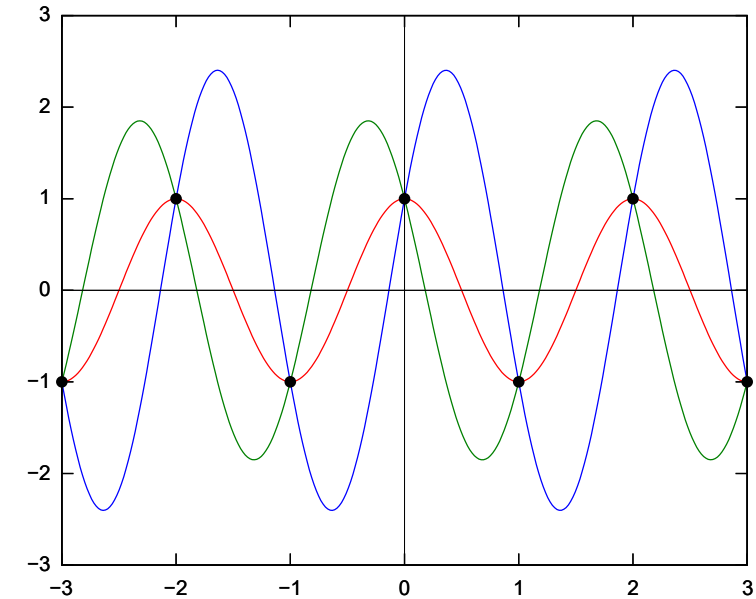
Aliasing and the critical frequency

■ The plot on the right shows

- A sequence of sampling points (alternating +1 and -1)
- Three sinusoids with identical frequency $f = \frac{1}{2} f_s$

■ Observations:

- All sinusoids match the sampling points (they are *aliases*)
- We cannot uniquely fit a sinusoid at frequency $f = \frac{1}{2} f_s$
- To determine the sine uniquely, we must sample with f strictly larger than $2f$: $f_s > 2f$



Source: [Wikimedia](#)

Tasks:

- What is the frequency f of the three sine curves?
- What is the relationship between f and ω in $\sin(\omega t)$?
- If f is the same for all three sines, then what changes?

Answers:

- The period is $T = 2$. Therefore $f = \frac{1}{T} = 0.5$
- $f = 2\pi\omega$
- The amplitude and the phase

Assignment

Sampling and aliasing are important concepts – yet they can be tricky at first. If you're feeling unsure, you're not alone. 😊

1. Read [this question](#) about **sampling near the Nyquist rate**
2. Study the five plots on the right and try to answer the OP's questions on your own.
3. Then review the provided answers and compare them with your reasoning.

